

Explainable Deep Learning for Deepfake Voice Detection: A Leakage-Free Re-evaluation and Cross-Dataset Study

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Abstract—We revisit a recently reported machine-learning approach to deepfake voice detection that claims 99.93% accuracy, and show that this figure stems from a leaky, small-scale evaluation rather than genuine detection ability. We merge three public corpora—ASVspoof 2021, a LibriSpeech/TTS set, and MLAAD v5—into a single balanced benchmark, and re-evaluate under a speaker-disjoint protocol using Equal Error Rate (EER). On this honest benchmark we reproduce the paper’s classical pipeline as a baseline (best 11.33% EER) and propose *AttentiveSpecCNN*, a compact convolutional network with a learnable temporal-attention pooling layer that both improves accuracy and yields built-in explanations via attention weights and Grad-CAM. The proposed model reaches 4.81% EER, a $\sim 58\%$ relative reduction over the baseline. Finally, a leave-one-source-out study shows that this in-domain success does *not* transfer to a fully unseen source, quantifying an important and often-overlooked generalisation gap.

Index Terms—audio deepfake detection, anti-spoofing, explainable AI, attention, EER, cross-dataset generalisation.

I. INTRODUCTION

Synthetic speech is now realistic enough to threaten voice authentication and enable fraud and misinformation, motivating reliable detectors. A recent study, “Unmasking the Fake” [1], reports 99.93% accuracy using Mel Frequency Cepstral Coefficients (MFCC) refined by a Gaussian-Naive-Bayes and Non-negative-Matrix-Factorisation “transfer” step feeding classical classifiers.

Such a figure is not comparable to properly evaluated anti-spoofing systems. It is obtained on a small, single Kaggle set using random k -fold cross-validation and synthetic minority oversampling applied *before* splitting, both of which leak information between train and test; the reported per-class metrics are internally inconsistent; and no EER—the field-standard metric—is given. We therefore treat the method as a baseline to reproduce rather than a target.

Our contributions are: (i) a unified, cleaned, class-balanced benchmark merged from three public corpora; (ii) a rigorous *speaker-disjoint* evaluation with EER; (iii) *AttentiveSpecCNN*, an explainable model that replaces the paper’s static feature engineering with a learnable attention mechanism and improves EER by $\sim 58\%$ relative over the reproduced baseline;

TABLE I

THE THREE MERGED SOURCE DATASETS. THE COMBINED CORPUS IS $\sim 97\%$ FAKE; A CLASS-BALANCED SUBSET (19,251 / 19,251) IS USED.

Source (local name)	Description	Real	Fake
ASVspoof 2021 (DF)	Telephone/codec deepfakes	16,977	441,894
Adarsh AD set	LibriSpeech real + TTS fakes	2,274	2,173
MLAAD v5	Multilingual TTS (fake only)	0	151,446
Merged	Unified manifest	19,251	595,513

and (iv) a cross-dataset study exposing the limits of in-domain performance.

II. DATASETS AND PREPROCESSING

We merge three public sources (Table I): the ASVspoof 2021 deepfake (DF) track [2], a set pairing LibriSpeech real speech with modern text-to-speech (TTS) fakes, and MLAAD v5 [3], a multilingual TTS corpus spanning roughly ninety generators. The combined corpus is $\sim 97\%$ fake, so we draw a class-balanced subset (19,251 real / 19,251 fake) whose fake half is spread evenly across all sources and generators to avoid a single-generator bias. Audio is decoded, converted to mono, resampled to 16 kHz, fixed to 4 s, and transformed to 80-band log-mel spectrograms.

III. METHOD

A. *AttentiveSpecCNN*

The proposed detector is a compact ($\approx 106\text{K}$ -parameter) 2-D CNN over log-mel spectrograms followed by a learnable temporal-attention pooling layer. Four convolutional blocks (channels 16/32/64/128, each 3×3 conv, batch norm, ReLU and 2×2 max-pool) produce a time–frequency feature map; the frequency axis is averaged and an attention layer computes a softmax weight per time frame, forming a context vector classified by a small multilayer perceptron. This attention layer is the key departure from [1]: rather than a fixed MFCC→GNB→NMF transform, it *learns* how much each moment of audio contributes to the decision. Table II lists the configuration.

TABLE II
AUDIO FRONT-END, MODEL, AND TRAINING CONFIGURATION.

Setting	Value
Sample rate / channels	16 kHz, mono
Clip length	4 s (64,000 samples); pad, else random (train) / center (eval) crop
Log-mel	80 bands, $n_{\text{fft}}=400$, hop=160 $\rightarrow$ 80×401
CNN backbone	4 blocks, 16/32/64/128, 3×3 conv+BN+ReLU+ 2×2 pool
Pooling	frequency mean, then temporal attention
Head	Linear 128 $\rightarrow$ 64, ReLU, dropout 0.3, Linear 64 $\rightarrow$ 2
Parameters	$\approx 106,000$
Optimiser	Adam, lr 10^{-3} , ReduceLROnPlateau ($\times 0.5$, patience 2)
Loss	cross-entropy, inverse-frequency class weights
Batch / epochs	64 / 15, best-val-EER checkpoint
Hardware	single NVIDIA RTX 3080

B. Explainability

Two saliency views come for free. The attention weights give a per-frame importance curve, and Grad-CAM over the final convolutional layer highlights the time–frequency regions that drove the predicted class (Fig. 2).

C. Classical baseline

We reproduce [1] with 40 MFCCs summarised by mean and standard deviation (an 80-D vector), standardised before classification. Random Forest, Logistic Regression, k -Nearest Neighbours and Gaussian Naive Bayes are evaluated on the raw MFCCs and on the paper’s GNB+NMF “transfer” features (GNB class probabilities concatenated with a 16-component NMF).

IV. EVALUATION PROTOCOL

Two choices make the numbers trustworthy. First, splits are *speaker-disjoint*: train, validation and test share no speaker or TTS source-reader, so a model cannot inflate its score by memorising voice identity, unlike random k -fold. Second, we report EER—the operating point where false accept and false reject rates are equal—alongside balanced accuracy and per-class F1, so the “predict-everything-fake” failure mode cannot hide behind raw accuracy.

V. RESULTS

A. In-domain detection

Table III and Fig. 1 report all methods on the same speaker-disjoint test set. The proposed model roughly halves the best classical baseline’s EER. Notably, the paper’s GNB+NMF “transfer” step does not help—it slightly worsens the strongest classical model—and the paper’s headline GNB is the weakest configuration.

B. Cross-dataset generalisation

To test whether the model detects transferable artifacts or dataset-specific cues, we train on two sources and test on the held-out third (Table IV, Fig. 3). Performance collapses to, or below, chance. Trained on ASVspoof (telephone/codec)

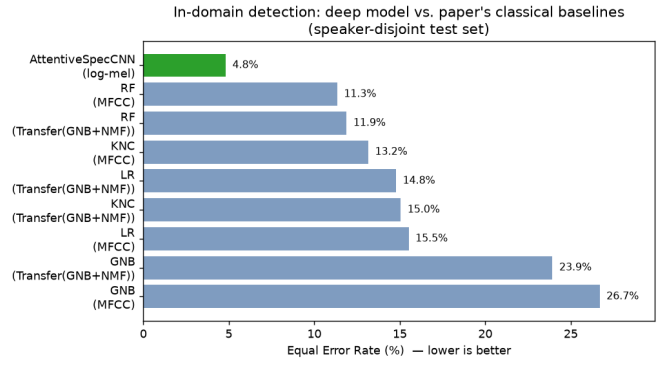


Fig. 1. Equal Error Rate by method; the proposed model roughly halves the best classical baseline’s EER.

and MLAAD (clean TTS), the model appears to associate clean audio with “fake” and mislabels the clean LibriSpeech real speech of dataset 2 (AUC 0.42, below chance). The strong in-domain result thus leans heavily on recording/domain signatures.

VI. DISCUSSION AND LIMITATIONS

The study supports a three-tier conclusion, each with protocol-correct numbers: the paper’s 99.93% accuracy is not reproducible or comparable; on a fair benchmark the proposed explainable model achieves 4.81% EER, a $\sim 58\%$ relative improvement over the reproduced baseline; and cross-dataset generalisation remains open, so in-domain metrics overstate real-world robustness. Limitations: MLAAD contributes only fake audio, so part of the separation reflects domain differences; the ASVspoof-holdout fold is real-starved (only $\sim 2.3\text{k}$ real clips to train on); and the cross-dataset numbers are single-run, fixed-threshold probes, for which ROC-AUC is the most robust signal.

VII. CONCLUSION

A leakage-free re-evaluation dispels an implausibly high published accuracy and shows that a compact, explainable attention CNN substantially improves honest in-domain detection. The cross-dataset collapse, however, underscores that robust generalisation—not headline accuracy—is the open problem in audio deepfake detection.

REFERENCES

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TABLE III
IN-DOMAIN DETECTION ON THE SPEAKER-DISJOINT TEST SET (LOWER EER IS BETTER).

Method	Features	EER (%)	Bal. Acc (%)	F1 _{fake} (%)	AUC
AttentiveSpecCNN (ours)	log-mel	4.81	93.84	93.49	0.99
Random Forest	MFCC	11.33	87.51	87.86	0.96
Random Forest	GNB+NMF transfer	11.89	87.60	87.83	0.95
k -Nearest Neighbours	MFCC	13.15	85.46	86.13	0.92
Logistic Regression	GNB+NMF transfer	14.78	84.73	84.09	0.92
k -Nearest Neighbours	GNB+NMF transfer	15.05	83.93	84.67	0.91
Logistic Regression	MFCC	15.52	84.33	84.17	0.91
Gaussian Naive Bayes	GNB+NMF transfer	23.93	76.72	75.77	0.87
Gaussian Naive Bayes (paper headline)	MFCC	26.69	73.05	71.75	0.81

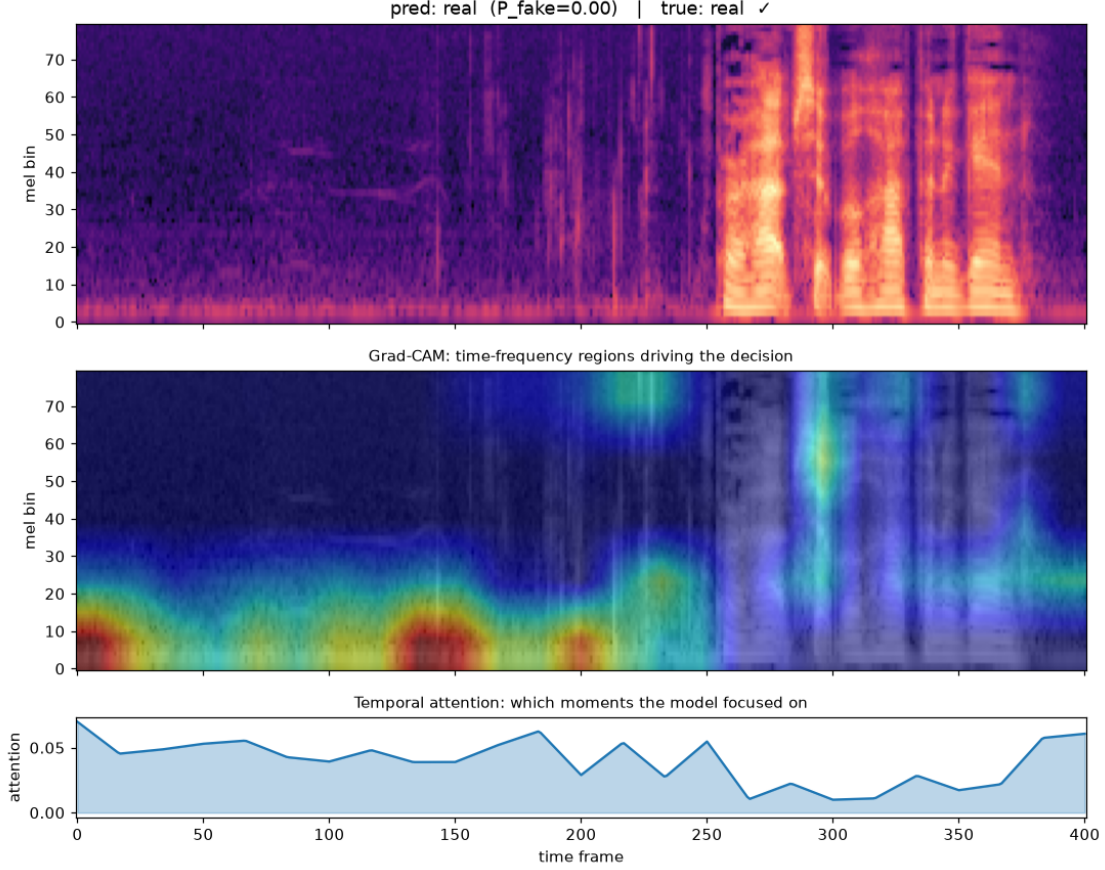


Fig. 2. Explanation for a correctly-identified real clip: log-mel spectrogram (top), Grad-CAM (middle), and temporal attention (bottom).

TABLE IV
LEAVE-ONE-SOURCE-OUT GENERALISATION. MLAAD HAS NO REAL AUDIO, SO ONLY THE FAKE-DETECTION RATE ON UNSEEN GENERATORS IS REPORTABLE.

Held-out source	Trained on	Metric	Result
(ref.) In-domain	all three	EER / AUC	4.81% / 0.99
ASVspoof 2021	ds2 + MLAAD	EER / AUC	44.9% / 0.56
dataset 2	ASV + MLAAD	EER / AUC	57.5% / 0.42
MLAAD	ASV + ds2	fake-recall	48.4%

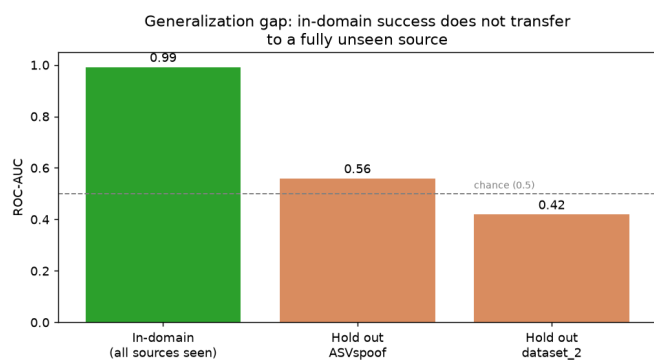


Fig. 3. In-domain vs. held-out ROC-AUC. Performance collapses to (or below) chance on a fully unseen source.